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# Step 1: Import required libraries
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Step 2: Upload an image
uploaded = files.upload()

# Step 3: Read the uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)

# Step 4: Convert image to RGB for displaying
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Step 5: Define Laplacian sharpening mask
laplacian_mask = [
    [0, -1, 0],
    [-1, 5, -1],
    [0, -1, 0]
]

# Step 6: Apply the Laplacian mask
sharpened = cv2.filter2D(image, -1,
    __import__('numpy').array(laplacian_mask, dtype='float32'))

# Step 7: Convert sharpened image to RGB
sharpened_rgb = cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB)

# Step 8: Display original and sharpened images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(sharpened_rgb)
plt.title("Sharpened Image - Laplacian Mask")
plt.axis("off")

plt.show()
```

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every-detail-of-a-sleek-modern-car-captured-in-close-up-photo.jpg(image/jpeg) - 35579 bytes, last modified: 9/1/2026 - 100% done

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Original Image



Sharpened Image - Laplacian Mask



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